

Nodejs bootcamp project

Introduction:

The bootcamp project is an initiative to introduce you to a holistic view of production grade nodejs and what it means to build production level software.

The premise is simple , build any 5 services from any provider but build it with the full architecture in place. You are required to make use of 5 separate HTTP verbs : GET , POST , PUT , PATCH , DELETE ETC.

Explicit Technology and Dependency Requirements:

- Nodejs version 21 and above
- Express js as a lightweight server framework
- Axios as an http wrapper for outbound services
- Jest for Unit testing
- Winston and Morgan for logging

Solution requirements:

- Your solution should be properly source controlled and display a clear branching strategy , each feature / function should be raised as a Pull Request to your main / master branch before approval and implementation to master
- Your solution should properly implement the following concepts as middlewares
 - Model driven request validation
 - JWT authentication and authorization
 - Custom error handling and validation
 - Custom route handlers
- Your solution should properly implement the following concepts within the technologies
 - Environmental variables
 - Configuration files
 - Logging
- Your solution should manage routes the following way
 - At Least 1 outbound API call via AXIOS
 - At Least 1 call using async / await syntax
 - Each service should have its own route
 - Each route should clearly separate API logic from functional logic
 - Please implement 5 different http verbs and demonstrate the appropriate knowledge of these verbs by using them correctly
- Your solution should implement examples of the following testing concepts:
 - All services should implement at least 1 unit test via Jest
 - At Least 2 Integration Tests in total
 - 1 E2E Test in total
- Your solution should implement the following build and deployment concepts:
 - Containerization